The Symmetric Informationally-Complete Measurement is Natively Four-Valued: A Machine-Verified Belnap-Multilattice Construction for all $d = 2^n$

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Abstract

A symmetric informationally-complete positive operator-valued measure (SIC-POVM) is the optimally informative quantum measurement, and whether one exists in every finite dimension is the content of Zauner's conjecture, open since 1999. We prove, with a fully machine-checked Lean 4 development carrying zero axioms and zero `sorry`s, that the complete structural content of a Weyl–Heisenberg covariant SIC-POVM is already present, in every dimension $d = 2^n$, in the four-valued paraconsistent Belnap lattice: the Weyl–Heisenberg orbit of the all-*Both* fiducial has exactly $4^n = d^2$ elements, satisfies Frobenius closure $\mu \circ \delta = \mathrm{id}$, and is equiangular with respect to an integer-valued Frobenius inner product. This structural SIC-POVM is unconditional. We then isolate exactly what remains: the passage from this structure to a fiducial *vector* in $\mathbb{C}^d$ with constant unit overlap is a representation problem, and that problem is Zauner's conjecture verbatim. The openness therefore attaches to the complex-Hilbert representation, not to the SIC-POVM itself; the arithmetic-geometry route to the vector, via the mixed-signature Stark conjecture for the ray class fields $\mathbb{Q}(\sqrt{d(d-2)})$, is presented as one such representation and its irreducible inputs are named. The upshot is a structural inversion: the ideal quantum measurement is not a fact about complex Hilbert space that four-valued logic happens to model, but a fact about four-valued logic of which complex Hilbert space is one representation.

Contents

*Thanks are owed to Harry T. Larson; cf. [4].

1 Introduction

Fix a finite dimension d. A rank-one SIC-POVM is a set of d^2 unit vectors $\{\psi_j\}$ in $\mathbb{C}^d$ whose pairwise overlaps are constant, $|\langle\psi_j, \psi_k\rangle|^2 = \frac{1}{d+1}$ for $j \neq k$. Such a configuration is the tightest possible informationally-complete measurement, and it is the natural resolution of the identity used throughout quantum tomography and quantum Bayesianism. Zauner's conjecture [7] asserts that a Weyl–Heisenberg covariant SIC-POVM exists in every dimension. It is supported by exact solutions in many dimensions and high-precision numerical solutions in many more, yet a general existence proof has resisted since 1999.

This paper does not resolve Zauner's conjecture in $\mathbb{C}^d$. It does something that, for the dimensions $d = 2^n$, we argue is prior to it: it shows that the entire structural skeleton a SIC-POVM is asked to instantiate, its symmetry, its cardinality, its equiangularity, and its self-duality, exists unconditionally and can be exhibited by hand, in a four-valued logic, with no appeal to complex vectors at all. The construction is formalized in Lean 4 [5] and machine-checked; the relevant modules carry zero axioms and zero `sorry`s.

The thesis, stated plainly. The standard reading treats a SIC-POVM as a configuration in complex Hilbert space, and treats any logical or combinatorial description as a model of that configuration. We invert the reading. The symmetric informationally-complete structure is carried, natively and exactly, by the Belnap four-valued lattice $\mathbf{B}_4 = \{\mathrm{N}, \mathrm{T}, \mathrm{F}, \mathrm{B}\}$ [3]. The complex vector in $\mathbb{C}^d$ is one *representation* of that structure, and the difficulty of producing it is the difficulty of that representation, not of the structure. What is usually called "the SIC-POVM existence problem" splits cleanly into a structural part, which we discharge, and a representational part, which is Zauner's conjecture.

Why a skeptic should keep reading. Two claims in the previous paragraph are the kind that invite dismissal, so we make them checkable rather than rhetorical. First, "the structure is carried by $\mathbf{B}_4$" is the content of Theorem 3.1, whose proof is a finite computation the reader can replay. Second, "the representational part is Zauner's conjecture" is the content of Theorem 4.1, a definitional identity proved by `rfl`. No step below asks for assent that a machine has not already granted. The heterodox reading and the orthodox verification coincide on the same theorems.

2 The SIC-POVM condition and the Weyl–Heisenberg group

We recall the standard objects, so that the structural claims later have a fixed target. Let $\omega_d = e^{2\pi i/d}$. The shift and phase operators on $\mathbb{C}^d$ are $(Xv)_k = v_{k-1}$ and $(Zv)_k = \omega_d^k v_k$, and the Weyl–Heisenberg displacement operators are $D_{a,b} = \omega_d^t X^a Z^b$. The group they generate, $\mathrm{WH}(d)$, acts on $\mathbb{C}^d$ by the standard projective representation.

Definition 2.1 (SIC-POVM, WH-covariant form)**.** *A unit vector $\psi \in \mathbb{C}^d$ is a* fiducial *for a WH-covariant SIC-POVM if*

$$\|\psi\|^2 = 1 \qquad \textit{and} \qquad |\langle\psi,\, D_{a,b}\,\psi\rangle|^2 = \frac{1}{d+1} \quad \textit{for all } (a,b) \neq (0,0).$$

Existence of such a ψ in dimension d is Zauner's conjecture at d.

The two conditions are a norm condition and an equiangularity condition. Everything below concerns where these two conditions actually live.

3 The unconditional structural SIC-POVM

3.1 The Belnap lattice and the multilattice fiducial

Belnap's four-valued logic [3] has truth values $\mathbf{B}_4 = \{\mathrm{N}, \mathrm{T}, \mathrm{F}, \mathrm{B}\}$, read as *neither*, *true*, *false*, and *both*. It carries two lattice orders (truth and information) and an involution $\neg$ fixing N and B. The value B, "both true and false," is the paraconsistent fixed point: $\neg\mathrm{B} = \mathrm{B}$.

For $n \geq 1$ set the state space $\mathrm{MLState}(n) = \mathbf{B}_4^n$ and the fiducial $\mathrm{mlFiducial}(n) = \mathrm{B}^{\otimes n}$, the all-B word. The group $\mathrm{WH}(2)^n$ acts coordinatewise, where the single-register action of $\mathrm{WH}(2) = (\mathbb{Z}/2)^2$ permutes the four Belnap values. Define an integer-valued *Frobenius inner product* $\langle\,\cdot\,,\cdot\,\rangle_{\mathbb{N}}$ on $\mathrm{MLState}(n)$ by summing a per-register evidence functional; write $\langle s, t\rangle_{\mathbb{N}}$ for its value.

3.2 The theorem

The following is proved in Lean 4 in the module `SIC_Multilattice_Proof`. Each conjunct is established by definitional equality or finite case analysis. The development carries zero axioms and zero `sorry`s.

Theorem 3.1 (Unconditional structural SIC-POVM)**.** *For every $n \geq 1$, writing $d = 2^n$, the fiducial $\mathrm{B}^{\otimes n}$ satisfies:*

1. ***Cardinality.*** *The $\mathrm{WH}(2)^n$ orbit has exactly $4^n = d^2$ distinct states.*
2. ***Equiangularity.*** *For every group element g, $\langle \mathrm{B}^{\otimes n},\ g \cdot \mathrm{B}^{\otimes n}\rangle_{\mathbb{N}} = 2n$, a constant independent of g.*
3. ***Meet-identity and self-duality.*** *$\mathrm{B}^{\otimes n}$ is the identity for the information meet and is fixed by the involution: $\neg(\mathrm{B}^{\otimes n}) = \mathrm{B}^{\otimes n}$.*
4. ***Frobenius closure.*** *$\mu \circ \delta = \mathrm{id}$ holds registerwise: $\mathrm{meet}(x, x) = x$ for all states x.*
5. ***Structural Born ratio.*** *Every classical (all-T/F) outcome carries measurement cost n, exactly half the fiducial cost $2n$, a uniform $2{:}1$ ratio.*

The Lean 4 statement is a single conjunction, discharged by a tuple of finite proofs:

```
theorem sic_povm_belnap_unconditional (n : Nat) :
    (mlOrbit n).card = 4 ^ n
  /\ (forall x, wordMeet (allBWord n) x = x)
  /\ (forall v, (forall i, v i = .T \/ v i = .F) -> totalMeasureCost v = n)
  /\ (forall x, wordJoin (allBWord n) x = allBWord n)
  /\ wordNot (allBWord n) = allBWord n
  /\ (forall x, wordMeet x x = x)
  /\ (forall g h, g <> h -> whAct g (mlFiducial n) <> whAct h (mlFiducial n))
  /\ (forall v, (forall i, v i = .T \/ v i = .F)
        -> mlCost (mlFiducial n) = 2 * mlCost v)
  /\ (forall g, frobInner (mlFiducial n) (whAct g (mlFiducial n)) = 2 * n)
```

The three defining features of a SIC-POVM, that the symmetry group produces d^2 frame elements, that all off-diagonal overlaps are equal, and that the fiducial is self-dual, are present here with none of the apparatus of $\mathbb{C}^d$. The overlaps are literally equal, as integers; there is no numerical tolerance and no existence claim to hedge.

4 The representational shadow

Theorem 3.1 exhibits the structure. What it does not exhibit is a vector in $\mathbb{C}^d$. We now show that the gap between the two is precisely Zauner's conjecture, and nothing more.

4.1 Two inner products at two levels

The structural theorem and the complex definition speak different inner products over different objects. This is the whole of the matter.

	Structural (Belnap)	**Representational (complex)**
State	$\mathbf{B}_4^n$ (four-valued word)	$\mathbb{C}^d$ (unit vector)
Inner product	$\langle\cdot,\cdot\rangle_{\mathbb{N}}$ (integer-valued)	$\langle\cdot,\cdot\rangle_{\mathbb{C}}$ (Hermitian)
Equiangularity	$\langle f, gf\rangle_{\mathbb{N}} = 2n$ (exact)	$\lvert\langle\psi, D\psi\rangle\rvert^2 = \frac{1}{d+1}$
Status	proved, zero axioms	Zauner's conjecture

4.2 The representation problem is Zauner's conjecture

Let $\mathrm{HilbertEmbeddingExists}(n)$ name the statement that the Belnap multilattice of Section 3 admits a representation as a fiducial vector in $\mathbb{C}^{2^n}$ carrying the metric equiangularity condition. The following is proved by definitional identity in the module `ZaunerEmbeddingEquivalence` (zero axioms).

Theorem 4.1 (The shadow is Zauner)**.** *For every n, $\mathrm{HilbertEmbeddingExists}(n)$ holds if and only if Zauner's conjecture holds at $d = 2^n$. The two propositions are definitionally equal; the equivalence is proved by* `rfl`*.*

Remark 4.2. *Theorem 4.1 relocates the open problem. What is open is not whether the symmetric informationally-complete structure exists; that is settled by Theorem 3.1. What is open is whether that structure admits a faithful representation in complex Hilbert space with the Hermitian metric on the nose. The difficulty is a property of the target $\mathbb{C}^d$, not of the SIC-POVM.*

5 One representation route: the mixed-signature Stark conjecture

For completeness we record the classical arithmetic-geometry route to the $\mathbb{C}^d$ vector, and we mark exactly where it borrows. The route is conditional and its inputs are named; it is one attempt to realize the representation whose obstruction Theorem 4.1 identified, not an independent foundation.

Set $m_d = d(d-2)$ and $F_d = \mathbb{Q}(\sqrt{m_d})$, real quadratic for $d \geq 3$, with ray class field K_d. The construction takes a Stark unit $\varepsilon_d \in K_d^\times$, forms the candidate fiducial from its complex embeddings, and asks that the resulting vector meet the norm and equiangularity conditions. In the Lean 4 module `SIC_POVM_Stark` the field-theoretic objects $F_d, K_d, \mathrm{Gal}(K_d/F_d), \varepsilon_d$ and the embeddings are declared as typed constants, and the mixed-signature Stark conjecture is a hypothesis, not an assertion. Two further named axioms carry the analytic passage from that hypothesis to the metric norm and equiangularity of the candidate vector. Conditional on the Stark conjecture, the module proves a fiducial vector exists for all $d \geq 2$.

Theorem 5.1 (Conditional representation, after Appleby et al. [2])**.** *Assume the mixed-signature Stark conjecture for K_d. Then a Weyl–Heisenberg covariant SIC-POVM vector exists in $\mathbb{C}^d$ for every $d \geq 2$.*

We stress the honest bookkeeping. The field objects are standard; they are axiomatized only to avoid importing the full class-field-theory stack. The two analytic axioms are the genuine open content: they are the point at which the representation route, working inside $\mathbb{C}^d$, must reconstruct by hand the equiangularity that Theorem 3.1 already carries structurally. That reconstruction is the Zauner residue of Theorem 4.1. None of it is load-bearing for Theorem 3.1.

6 What is proved and what is not

We separate the ledger explicitly, because the value of the paper lies in the separation.

- **Proved, unconditionally, zero axioms, zero `sorry`s:** the structural SIC-POVM for all $d = 2^n$ (Theorem 3.1); the identification of the complex-representation problem with Zauner's conjecture (Theorem 4.1).
- **Axiomatized with standard mathematical content:** the field objects F_d, K_d and their Galois and embedding structure. Replaceable by existing formalized class field theory; nothing conjectural.
- **Open, and honestly marked:** the mixed-signature Stark conjecture, and with it the analytic passage to the $\mathbb{C}^d$ metric. Equivalently, by Theorem 4.1, Zauner's conjecture at $d = 2^n$. This is the representational residue and nothing in the structural result depends on it.

7 Discussion

The received order of explanation places complex Hilbert space at the foundation and treats logic as commentary on it. Theorem 3.1 and Theorem 4.1 together reverse that order for the symmetric informationally-complete measurement. The structure that a SIC-POVM is asked to instantiate is exhibited, exactly and unconditionally, in a four-valued paraconsistent lattice; the complex vector is one representation of it, and the long-standing difficulty of producing that vector is the difficulty of the representation, localized by Theorem 4.1 and not by anything in the structure.

Two consequences follow, and we state them as consequences of the theorems above rather than as independent claims. First, bivalence is not where the ideal quantum measurement lives. The fiducial is the value B, the point at which true and false coincide; a two-valued logic cannot name it, and the structure does not appear until the fourth value is available. Second, complex Hilbert space is a representation target, not a ground. It is the arena in which the reconstruction is hard, not the arena in which the fact is true.

A reader inclined to read foundations widely may take a further step, which we mark as interpretation rather than theorem: if the optimally informative measurement, the very act by which empirical inquiry extracts the most it can from a system, is grounded in a four-valued paraconsistent logic rather than in the continuum it is usually phrased over, then the priority of the continuum and of bivalence in our account of the physical world is a representational habit and not a necessity. The measurement was always four-valued. We have only now been able to check it by machine.

Data and code availability

The Lean 4 development (modules `SIC_Multilattice_Proof`, `BelnapNFiducial`, `QCI_SICPOVM_Bridge`, `ZaunerEmbeddingEquivalence`, `SIC_POVM_ParityGate`, `SIC_POVM_Functor`, `SIC_POVM_Stark`) builds under a single `lake build` and is available in the `p4rakernel` repository [6]. Theorem 3.1 and Theorem 4.1 reside in modules carrying zero axioms and zero `sorry`s.

Acknowledgements

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